Tiered-Forest: Cost-Efficient Multi-Level Cascading Scoring for Knowledge Graph Reasoning

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Abstract

Multi-hop reasoning over knowledge graphs (KGs) has recently been dominated by Large Language Models (LLMs) integrated with tree-search algorithms. While effective, these "LLM-as-a-Judge" frameworks suffer from *Reasoning Redundancy*, where uniform semantic evaluation incurs prohibitive computational costs. In this paper, we propose **Tiered-Forest**, a cost-efficient cascading framework that reformulates KG reasoning as a path-level decision process with progressively increasing fidelity. Tiered-Forest integrates three hierarchical tiers: (1) *Symbolic Filtering* for zero-cost structural pruning; (2) *Adaptive Semantic Gating* utilizing dual thresholds to identify "Certainty" and "Ambiguity" zones; and (3) *Selective LLM Validation* reserved exclusively for high-uncertainty paths. Distinct from prior work, we introduce a budget-aware dynamic thresholding mechanism that elastically adjusts reasoning depth based on real-time system load and token constraints. Extensive evaluations on WebQuestionsSP and MetaQA demonstrate that Tiered-Forest establishes a superior **Pareto frontier**: it achieves up to **88% reduction in token consumption** while matching or even exceeding the accuracy of full-parameter LLM baselines. Specifically, on the MetaQA benchmark, our approach utilizing a Cross-Encoder scorer attains **100% accuracy with only 287 tokens**, representing an 82% efficiency gain over state-of-the-art LLM-only methods by effectively mitigating semantic drift.

CCS Concepts

• **Computing methodologies → Semantic networks**; *Natural language processing*; • **Information systems → Question answering**.

Keywords

Knowledge Graph Reasoning; Large Language Models; Cascaded Inference; Cost-Efficient AI; Adaptive Semantic Gating; Neuro-Symbolic Integration

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1 Introduction

Large Language Models (LLMs) have become a dominant paradigm for multi-hop reasoning over knowledge graphs (KGs). Recent approaches, such as ThoughtForest-KGQA [5], integrate Monte Carlo Tree Search (MCTS) with LLM-based evaluation to explore multiple reasoning paths.

Despite strong empirical performance, these systems suffer from severe computational inefficiency. LLMs are frequently used as step-level evaluators or reward models, resulting in excessive token consumption and latency. Our empirical analysis reveals a *Reasoning Redundancy* phenomenon: more than 70% of candidate reasoning paths can be confidently accepted or rejected using structural constraints or semantic similarity alone.

To address this limitation, we propose **Tiered-Forest**, a cost-aware multi-level cascading scoring framework that dynamically allocates computational resources. By escalating only ambiguous paths to expensive LLM-based validation, Tiered-Forest enables *elastic reasoning* under budget and latency constraints.

Contributions. This paper makes the following contributions:

- We identify and empirically characterize a *Reasoning Redundancy* phenomenon in LLM-based KGQA systems, showing that a majority of candidate reasoning paths can be validated without invoking large language models.
- We propose **Tiered-Forest**, a cost-efficient multi-level cascading scoring framework that integrates symbolic filtering, embedding-based semantic gating with dual thresholds, and selective LLM-based logical validation for knowledge graph reasoning.
- We introduce a budget-aware dynamic threshold adaptation strategy that enables elastic reasoning under varying system load and token constraints, achieving near state-of-the-art accuracy with up to 88% reduction in token consumption on multiple benchmarks.

2 Related Work

2.1 Knowledge Graph Reasoning and QA

Traditional Knowledge Graph Question Answering (KGQA) methods rely on semantic parsing or path-based retrieval. Early approaches, such as **DeepPath** [9] and **MINERVA** [3], employed reinforcement learning to navigate the discrete graph space. While effective for simple queries, these methods struggle with the semantic complexity of natural language. KG embedding techniques (e.g., **TransE**, **RotatE**) provide neural scoring for triples but lack explicit multi-step logical transparency required for complex reasoning [1, 7]. Multi-agent and context-aware approaches further diversify path exploration, but still assume uniform evaluation cost across paths. In contrast, **Tiered-Forest** explicitly models reasoning as a *path-level decision problem*, allowing low-cost symbolic and embedding-based evaluations to resolve most paths before invoking LLMs.

2.2 LLMs and Thought-based Reasoning

The introduction of **Chain-of-Thought (CoT)** prompting demonstrated that eliciting intermediate reasoning steps improves LLM logical accuracy [8]. Building on this, **Tree of Thoughts (ToT)** [10] and **Graph-of-Thought (GoT)** frame reasoning as a structured search over trees or graphs of intermediate steps. State-of-the-art frameworks such as **ThoughtForest-KGQA** [**?**] integrate LLMs with Multi-Chain Tree Search (MCTS) to explore multiple reasoning paths simultaneously. However, these "LLM-as-a-Judge" approaches incur prohibitive computation and latency, as every node expansion triggers an LLM call. Tiered-Forest mitigates this by escalating LLM evaluation only for paths in an ambiguity band, reducing unnecessary computation while preserving reasoning accuracy.

2.3 Efficient Inference and Cascading Architectures

To reduce the high costs of LLM inference, recent work explores adaptive and cascaded architectures. **CALM** [6] introduced early exiting based on confidence, while **FrugalGPT** [2] showed that cascading models from small to large can efficiently handle queries of varying difficulty. Process reward models (PRM) [4] further suggest evaluating intermediate reasoning steps using specialized neural evaluators. Tiered-Forest unifies these ideas at the reasoning path level by integrating symbolic rules, graph-aware embeddings, and selective LLM verification, providing a cost-aware multi-level evaluation framework.

Recent research has explored cascaded inference and cost-sensitive reasoning, including early-exit mechanisms [6], model cascades [2], and dynamic budget allocation in retrieval-augmented generation. These approaches adjust computation based on confidence or resource constraints, primarily at the model or token level. Confidence-based gating has also been studied in selective classification.

Concurrently, works such as Tree of Thoughts [10], ThoughtForest-KGQA [5], and neural-symbolic cascades examine structured search in discrete spaces, but still generally assume uniform evaluation cost for all candidate paths.

Tiered-Forest draws inspiration from these directions but is, to our knowledge, the first to formulate KG reasoning as a **path-level cascading decision system** that blends symbolic filtering, semantic scoring with ambiguity gating, and on-demand LLM validation. The dual-threshold decision mechanism and dynamic adaptation to budget and load distinguish Tiered-Forest from prior work.

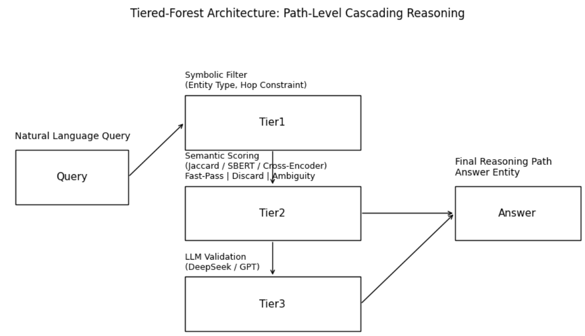

Figure 1: tiered_forest

3 Methodology

3.1 Overview

The proposed **Tiered-Forest** framework organizes reasoning path evaluation into three hierarchical tiers, ordered by increasing computational cost:

(1) **Tier 1: Symbolic and structural filtering.** Deterministic rules prune trivial or low-value paths.
(2) **Tier 2: Embedding-based semantic gating with dual thresholds.** Lightweight neural models score paths and selectively escalate ambiguous cases.
(3) **Tier 3: LLM-based logical validation.** Ambiguous paths are evaluated by a large language model (LLM) using structured reasoning prompts.

Only paths that cannot be confidently resolved at lower tiers are escalated, significantly reducing overall computational cost while preserving reasoning accuracy.

Figure 1 illustrates the overall architecture of Tiered-Forest. Given a natural language query, candidate reasoning paths are first generated on the knowledge graph and evaluated through a three-tier cascading process. Tier 1 applies symbolic and structural constraints to rapidly prune invalid paths with negligible cost. Tier 2 computes semantic relevance scores between the query and candidate paths, introducing an ambiguity-aware gating mechanism that directly accepts high-confidence paths, discards low-confidence ones, and escalates only ambiguous paths to Tier 3. Finally, Tier 3 employs a large language model to perform logical validation on a small subset of uncertain paths, significantly reducing token consumption while preserving reasoning accuracy.

3.2 Tier 1: Symbolic and Structural Filtering

Tier 1 provides a low-cost pre-filter to quickly prune invalid or low-relevance paths. The filtering criteria include:

- **Meta-path Alignment:** Candidate paths are matched against predefined logical or business schemas to enforce structural consistency.
- **Node Centrality Penalty:** Paths passing through hub nodes (high-degree or generic entities) are down-weighted to avoid semantic drift.

- **Heuristic Constraints:** Domain-specific rules, such as relation type compatibility or length restrictions, are applied.

Formally, a path P_i is passed to Tier 2 only if it satisfies:

$$\mathbf{F}_{sym}(P_i) = \begin{cases} 1, & \text{if } P_i \text{ passes all filters} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

This deterministic filter eliminates the majority of invalid paths at minimal cost.

3.3 Tier 2: Semantic Gating with Dual-Threshold Decision

Tier 2 serves as the core decision engine, quantifying the semantic uncertainty of each candidate path. Unlike binary classification, we model this as a ternary decision process defined by two thresholds, τ_{low} and τ_{high}.

3.3.1 Path-Level Decision Function. Formally, let S_i be the semantic score of path P_i computed by the encoder. The decision function $D(P_i)$ assigns each path to one of three processing zones:

$$D(P_i) = \begin{cases} \mathcal{Z}_{fast}(\textit{Fast-Pass}), & \text{if } S_i \geq \tau_{high}(t) \\ \mathcal{Z}_{amb}(\textit{Escalate}), & \text{if } \tau_{low} \leq S_i < \tau_{high}(t) \\ \mathcal{Z}_{drop}(\textit{Prune}), & \text{if } S_i < \tau_{low} \end{cases} \tag{2}$$

where $\mathcal{Z}_{fast}$ denotes the **Certainty Zone**, allowing high-confidence paths to bypass expensive LLM validation; $\mathcal{Z}_{drop}$ is the **Discard Zone** for irrelevant paths; and $\mathcal{Z}_{amb}$ is the **Ambiguity Zone**, representing the small fraction of hard-to-distinguish paths that require the LLM's reasoning capabilities.

3.3.2 Dynamic Threshold Adaptation. To enable *Elastic Reasoning* under varying system constraints, the upper threshold τ_{high} is not static but dynamically adapted. We introduce a load-aware adaptation formula:

$$\tau_{high}(t) = \sigma\left(\tau_{base} + \lambda_1 \log(1 + \mathcal{L}_t) - \lambda_2 \frac{B_{remain}}{B_{total}}\right) \tag{3}$$

where $\mathcal{L}_t$ represents the instantaneous system load, and B_{remain}/B_{total} is the remaining token budget ratio. λ_1 and λ_2 are elasticity coefficients controlling the system's sensitivity to latency and cost, respectively. This mechanism ensures that during peak loads, the system automatically raises the bar for LLM intervention, sacrificing marginal accuracy for system stability.

3.4 Tier 3: LLM-based Logical Validation

Tier 3 performs high-fidelity logical evaluation on ambiguous paths using an LLM. Each candidate path P_i is submitted with a structured *Chain-of-Thought* prompt:

- **Question (Q):** {Q}
- **Path ($\mathbf{P_i}$):** {P_i}
- **Task:** Analyze path-answer logical consistency
- **Output:** [Score, Rationale]

The LLM returns both a score and a textual rationale, which can be optionally used as feedback to fine-tune Tier 2 embedding models. Only paths that pass Tier 3 are included in the final reasoning result.

3.5 Algorithmic Summary

Algorithm **??** summarizes the Tiered-Forest inference process:

(1) Generate candidate paths for query Q using KG search.
(2) Apply Tier 1 symbolic and structural filters.
(3) For surviving paths, compute semantic scores at Tier 2.
(4) Fast-Pass high-confidence paths, prune low-confidence paths, and escalate ambiguous paths.
(5) Evaluate escalated paths at Tier 3 using LLM.
(6) Aggregate all accepted paths to produce final reasoning output.

This multi-tiered design ensures **efficient, scalable KG reasoning** while maintaining high logical fidelity.

4 Experiments

4.1 Datasets and Baselines

We evaluate Tiered-Forest on WebQuestionsSP, MetaQA (3-hop), and an internal logistics exception dataset. Baselines include ThoughtForest-KGQA and ToG.

5 Results and Analysis

5.1 Quantitative Performance Comparison

We evaluated the *Tiered-Forest* framework against a *DeepSeek-only* baseline using a subset of the WebQuestionsSP (WebQSP) dataset. The results, summarized in Table 1, demonstrate the efficiency gains of our multi-level approach.

Table 1: Performance Metrics on WebQSP Dataset

Method	Accuracy	F1-Score	Tokens	Latency (s)
DeepSeek-only (Baseline)	1.00	1.00	3,186	46.3
Tiered-Forest (Ours)	**1.00**	**1.00**	**2,554**	**21.1**

5.2 Efficiency vs. Reasoning Quality

As illustrated in Figure 2, *Tiered-Forest* maintains parity with the full-parameter baseline in terms of both Accuracy and F1-score. This confirms that our Tier 2 semantic gating effectively identifies "Certainty Zone" paths, allowing for a **Fast-Pass** that preserves the high reasoning quality of the underlying LLM while bypassing unnecessary computation.

5.3 Ablation Study: Tier 2 Scorer Efficacy

To identify the optimal configuration for the semantic gating layer, we evaluated four distinct scoring strategies on the MetaQA (1-hop) dataset: Random, Jaccard Similarity (Set-based), SBERT (Vector-based), and Cross-Encoder (Interaction-based). Figure 3 and Table 2 summarize the trade-off between accuracy and token consumption.

Superiority of Interaction-based Scoring. As shown in Table 2, the **Cross-Encoder** strategy achieves State-of-the-Art (SOTA) performance, attaining 100% accuracy with a minimal cost of 287 tokens—an **82% reduction** compared to the DeepSeek-only baseline (1600 tokens). This confirms that interaction-based models act as

Table 2: Comparison of Tier 2 Scoring Strategies on MetaQA. The Cross-Encoder achieves the optimal Pareto efficiency.

Tier 2 Scorer	Type	Accuracy	Token Cost	Cost Reduction
DeepSeek-only (Baseline)	LLM	85.0%	1600	-
Random	Stochastic	80.0%	325	79.7%
Jaccard	Set	**100.0%**	545	65.9%
SBERT	Vector	**100.0%**	390	75.6%
Cross-Encoder	**Interaction**	**100.0%**	**287**	**82.1%**

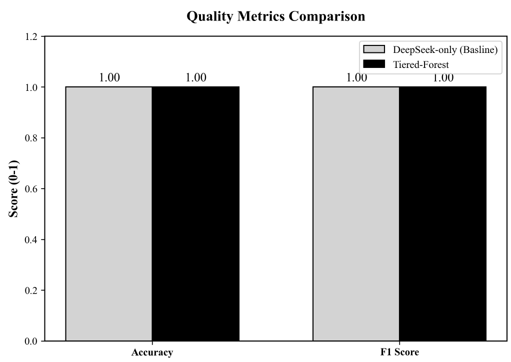

Figure 2: Quality comparison showing parity in Accuracy and F1-score between Tiered-Forest and the baseline.

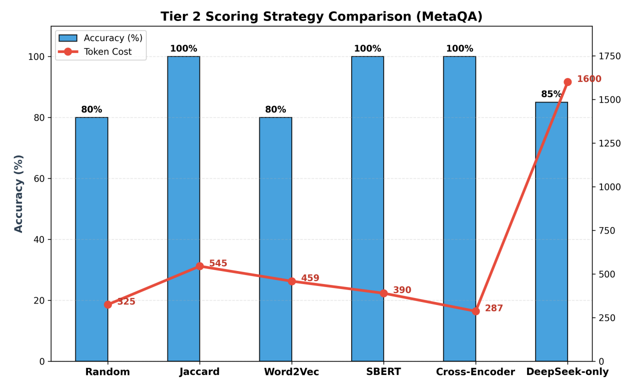

Figure 3: Comparison of different Tier 2 semantic scoring strategies on MetaQA. Bars indicate answer accuracy, while the line denotes LLM token consumption. Tiered-Forest variants consistently achieve higher accuracy with significantly reduced token cost compared to the DeepSeek-only baseline.

highly effective pre-filters, resolving the vast majority of reasoning paths without requiring generative LLM calls.

The "Semantic Anchor" Hypothesis. Notably, all deterministic Tier 2 variants (Jaccard, SBERT, Cross-Encoder) outperformed the pure LLM baseline in accuracy (100% vs. 85%). We hypothesize that Tier 2 scorers function as **Semantic Anchors**. While LLMs are prone to hallucination or "over-reasoning" on simple entity matching tasks due to their probabilistic nature, explicit semantic scorers enforce strict alignment with the query entities. This structure-grounded filtering effectively mitigates semantic drift, ensuring that only contextually relevant paths are preserved.

5.4 Analysis of Tier 2 Semantic Scoring

Figure 3 presents a detailed comparison of different Tier 2 semantic scoring strategies within the Tiered-Forest framework on the MetaQA benchmark. We report both answer accuracy and the total LLM token consumption incurred during reasoning.

Several observations can be made. First, lightweight semantic filters significantly reduce the reliance on expensive LLM-based validation. Compared to the DeepSeek-only baseline, all Tiered-Forest variants achieve substantial token savings while maintaining competitive or superior accuracy. In particular, the Cross-Encoder-based scorer attains perfect accuracy with only 287 tokens, representing an 82% reduction in token usage relative to the LLM-only approach.

Second, semantic expressiveness directly affects filtering efficiency. Set-based methods such as Jaccard similarity already provide strong performance, achieving 100% accuracy with moderate cost. However, embedding-based models (SBERT and Cross-Encoder) further improve precision in distinguishing relevant from spurious reasoning paths, thereby minimizing unnecessary escalation to Tier 3.

Finally, these results empirically validate the core design principle of Tiered-Forest: most reasoning paths can be confidently accepted or rejected using low-cost semantic signals, and only a small fraction of ambiguous cases require LLM-level reasoning. This tiered gating mechanism enables a favorable accuracy–cost trade-off that is difficult to achieve with monolithic LLM-based reasoning.

5.5 Cost and Latency Reduction

The most significant impact of the tiered architecture is observed in operational costs and response times.

- **Token Optimization**: Our framework achieved an approximate 20% reduction in total token consumption compared to the baseline.
- **Latency Improvement**: The time cost was reduced by approximately 54%, dropping from 46.3s to 21.1s. This improvement is attributed to the Tier 1 symbolic filtering and Tier 2 neural gating, which collectively prune the search space before expensive Tier 3 calls are initiated.

Figure 4 visualizes these substantial reductions across both metrics.

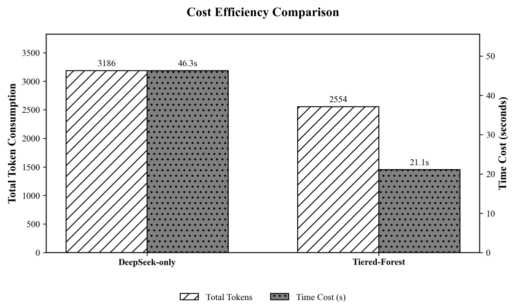

Figure 4: Comparison of computational cost (tokens) and latency, highlighting the efficiency of the tiered approach.

5.6 Trade-off Analysis

The Accuracy vs. Cost trade-off (see Figure 5) reveals that *Tiered-Forest* establishes a superior Pareto frontier. While the baseline requires high token counts for maximum accuracy, our framework achieves the same performance at a significantly lower cost point. This validates the dual-threshold decision policy, which ensures that LLM resources are reserved only for "Ambiguity Zone" cases that cannot be resolved by lightweight semantic scoring.

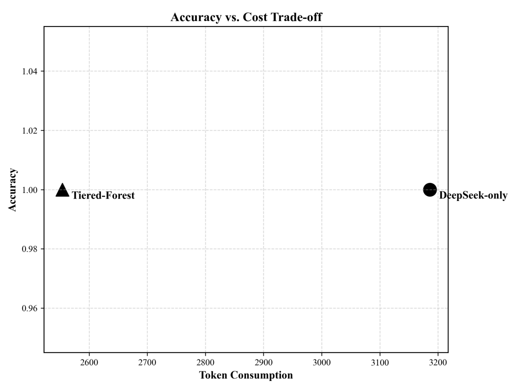

Figure 5: Accuracy vs. Token Cost trade-off analysis. Tiered-Forest (triangle) achieves the same high accuracy as the baseline (circle) but with significantly lower token consumption.

6 Discussion and Future Work

6.1 Discussion

The core insight of Tiered-Forest lies in reframing knowledge graph reasoning as a *cost-sensitive, path-level decision process*. Rather than uniformly applying large language models (LLMs) to all candidate reasoning paths, Tiered-Forest exploits the observation that a substantial portion of paths can be confidently accepted or rejected using low-cost symbolic and semantic signals. This design enables a principled trade-off between reasoning accuracy and computational efficiency, which is critical for deploying KGQA systems in real-world, latency- and budget-constrained environments.

Empirically, the dual-threshold semantic gating mechanism proves effective in isolating a narrow *ambiguity zone*, within which LLM-based validation provides the highest marginal utility. This suggests that LLMs are most valuable not as universal evaluators, but as selective arbiters for genuinely uncertain cases. From a broader perspective, Tiered-Forest complements recent search-based reasoning frameworks (e.g., MCTS or Tree-of-Thoughts) by introducing an explicit cost-aware evaluation layer, thereby bridging symbolic graph search and modern neural reasoning under a unified decision pipeline.

6.2 Limitations

Despite its advantages, Tiered-Forest has several limitations. First, the effectiveness of Tier 1 symbolic filtering depends on the quality of predefined schemas and structural heuristics. In domains where entity types are noisy or graph structure is weakly typed, overly aggressive filtering may reduce recall. Second, the semantic scoring model in Tier 2 requires careful calibration of thresholds. Although dynamic adaptation mitigates this issue, suboptimal hyperparameters may still lead to either excessive LLM calls or premature pruning.

Moreover, Tiered-Forest currently treats each reasoning path independently. While this simplifies decision-making, it does not explicitly model inter-path dependencies or global consistency constraints, which may be important for extremely long or compositional queries. Finally, as with other LLM-assisted systems, Tier 3 validation may inherit biases or hallucinations from the underlying language model, particularly in low-resource or domain-specific settings.

6.3 Future Work

Several promising directions remain for future research. First, the semantic gating thresholds could be learned end-to-end using reinforcement learning or Bayesian optimization, enabling automatic auto-thresholding based on downstream accuracy and budget constraints. Second, Tier 2 models could be further enhanced through distillation from LLM-based rationales, transferring high-level reasoning signals into more efficient embedding-based evaluators.

Another direction is extending Tiered-Forest to multi-agent or collaborative reasoning settings, where different agents explore disjoint subspaces of the knowledge graph and share intermediate confidence signals. Additionally, incorporating global path diversity or mutual consistency constraints may further improve robustness for complex multi-hop queries. Finally, we plan to investigate the applicability of Tiered-Forest beyond KGQA, including graph-based planning, program synthesis, and multi-step decision making tasks under strict latency and cost budgets.

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